Project Report: Hybrid NFC-Based University Attendance System

Report of 24th March:

1. Introduction

This project demonstrates the development of a **Hybrid NFC-Based University Attendance System** using **Flutter (frontend)**, **Google Apps Script (backend)**, and **Android NFC (HCE - Host Card Emulation)**.

The system enables:

- Student registration with photo
- Data storage in Google Sheets
- NFC-based student identification
- Attendance tracking via API

2. System Architecture

- **Frontend**
  - Built using **Flutter**
  - Runs on web (localhost:58552) and Android
  - Includes:
    - Student Signup Page
    - Login Page
    - Student ID & Wallet features
- **Backend**
  - **Google Apps Script (GAS)**
  - Acts as REST API (doGet, doPost)
  - Stores data in:
    - students sheet
    - attendance_raw2 sheet

🔹 **NFC Layer**

- Implemented using **Android HCE (HostApduService)**
- Sends student ID as APDU response

3. Student Registration Module

UI (Flutter)

From the screenshot:

- A complete **Student Sign Up form** is implemented with:
  - Student ID
  - Full Name
  - Department
  - Batch
  - Session
  - Email
  - Phone
  - Username
  - Password
  - Photo Upload

Issue Observed

Registration error: ClientException: Failed to fetch

Cause

- Backend API not accessible OR
- Permissions not granted in Google Apps Script

4. Google Apps Script Backend

Features Implemented

From screenshots:

- doGet(e) → handles read/mark actions

- doPost(e) → handles:
    - registerStudent
    - approveStudent
- Spreadsheet integration:

```
const SPREADSHEET_ID = "...";
const STUDENTS_SHEET_NAME = "students";
```

Authorization Issue

Error:

Exception: You do not have permission to call DriveApp.getRootFolder

Solution Applied

Steps followed:

1. Run doGet manually
2. Click **Review Permissions**
3. Accept warning:
    - "Google hasn't verified this app"
4. Click:
   Advanced → Go to project (unsafe)
5. Allow access

5. API Functionality

🔹 **Actions Supported**

```
if (action === "getPendingStudents")
if (action === "approveStudent")
return markAttendance(e);
```

🔹 **Student Approval Logic**

- Searches student ID in sheet
- Updates status to "Approved"

6. Android NFC Implementation

File: MyHostApduService.kt

Key logic:

```
override fun processCommandApdu(...) {
   val studentId = "20230045"
   return studentId.toByteArray(Charsets.UTF_8) + OK
}
```

Function

- Simulates NFC card
- Sends student ID when scanned

7. Android Configuration

AndroidManifest.xml

Includes:

- NFC permission:

```
<uses-permission android:name="android.permission.NFC"/>
```

- HCE service:

```
<service android:name=".MyHostApduService"
android:permission="android.permission.BIND_NFC_SERVICE">
```

apduservice.xml

Defines:

- AID group
- NFC service configuration

8. Development Environment

Tools Used

- VS Code
- Flutter SDK
- Android Studio (Gradle + Kotlin)
- Google Apps Script Editor

Project Structure

- lib/ → Flutter UI & logic
- android/ → NFC + native config
- assets/ → images
- scripts → GAS backend

9. Issues & Fixes

Issue	Cause	Solution
Failed to fetch	API not authorized	Grant permissions
DriveApp permission error	Missing OAuth consent	Run script manually
Google verification warning	Unverified app	Proceed via Advanced
NFC not working (potential)	Incorrect AID/service config	Fixed via apduservice.xml

10. Conclusion

The system successfully integrates:

- **Flutter UI**
- **Google Sheets backend via Apps Script**
- **Android NFC (HCE)**

Achievements

- Full student registration system
- API-based backend
- NFC-based identity simulation

Future Improvements

- Add secure authentication (JWT/Firebase)
- Improve error handling

- Deploy backend properly (not temporary URL)
- Add real NFC reader integration

11. Summary

This project demonstrates a **complete end-to-end attendance system**, combining:

- Mobile development
- Cloud scripting
- NFC technology

It is a strong foundation for a **real-world smart university system**.